

# Operational Equity Absorption and the Cross Section of Stock Returns

I. M. Harking

October 20, 2025

## Abstract

This paper studies the asset pricing implications of Operational Equity Absorption (OEA), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023a\)](#). A value-weighted long/short trading strategy based on OEA achieves an annualized gross (net) Sharpe ratio of 0.48 (0.42), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 32 (30) bps/month with a t-statistic of 3.49 (3.37), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Growth in book equity, Net Payout Yield, Share issuance (1 year), Share issuance (5 year), Change in equity to assets, Net equity financing) is 28 bps/month with a t-statistic of 3.22.

# 1 Introduction

Financial markets exhibit persistent deviations from perfect efficiency, with certain firm characteristics predicting future stock returns in ways that challenge traditional asset pricing models. While the efficient market hypothesis suggests that prices should rapidly incorporate all available information, mounting evidence reveals that information diffuses gradually through markets, creating predictable patterns in the cross-section of returns. Despite extensive research documenting hundreds of return predictors, we still lack complete understanding of how operational decisions reflected in firms' financial statements translate into stock returns. This paper examines Operational Equity Absorption (OEA), a novel signal that captures firms' use of equity capital in their operating activities, and demonstrates its robust ability to predict future returns.

We hypothesize that OEA predicts stock returns through the mechanism of slow information diffusion, building on seminal work showing that investors exhibit limited attention and process information gradually [Hong and Stein \(1999\)](#); [DellaVigna and Pollet \(2009\)](#); [Hirshleifer et al. \(2011\)](#). When firms absorb equity into operations through activities like stock-based compensation or equity-financed acquisitions, this information enters financial statements but diffuses slowly through the market as investors with limited attention fail to immediately recognize its implications for future profitability. The gradual incorporation of this information creates initial underreaction, followed by predictable return continuation as prices slowly adjust to reflect the full implications of operational equity changes [Hong et al. \(2000a\)](#); [Cohen and Frazzini \(2008\)](#).

The slow diffusion mechanism predicts stronger return predictability where information frictions are most severe. Small firms with low analyst coverage face greater barriers to information transmission, as fewer sophisticated intermediaries process and disseminate relevant signals [Hong et al. \(2000b\)](#); [Zhang \(2006\)](#). Similarly, firms

with lower institutional ownership or less liquid stocks experience slower price adjustment to new information, as these stocks attract less attention from informed traders who might otherwise accelerate price discovery [Peng and Xiong \(2006\)](#); [Barberis and Shleifer \(2003\)](#). These cross-sectional predictions provide testable implications that distinguish the slow diffusion hypothesis from alternative explanations based on risk or market microstructure.

The economic magnitude of OEA’s predictive power should manifest most strongly in environments where attention constraints bind. During periods of high market-wide information flow or when competing signals overwhelm investors’ processing capacity, we expect OEA’s return predictability to intensify [Hirshleifer et al. \(2009\)](#); [Kacperczyk et al. \(2016\)](#). This framework suggests that OEA captures a fundamental inefficiency in how markets process complex accounting information, with prices adjusting slowly as information gradually reaches and is processed by the marginal investor.

Our empirical analysis reveals that OEA strongly predicts cross-sectional stock returns, with a value-weighted long-short strategy achieving an annualized gross Sharpe ratio of 0.48 and delivering monthly average excess returns of 34 basis points with a t-statistic of 3.74. The strategy’s performance remains robust after controlling for established risk factors, generating a monthly alpha of 32 basis points (t-statistic = 3.49) relative to the Fama-French six-factor model that includes momentum. These results demonstrate both statistical significance and economic importance, with the strategy’s gross Sharpe ratio exceeding 87% of documented anomalies in the factor zoo.

The predictive power of OEA persists across various portfolio construction methodologies and after accounting for transaction costs. Net of trading costs, the strategy maintains a Sharpe ratio of 0.42 and delivers monthly net returns of 30 basis points (t-statistic = 3.33), with generalized net alphas remaining positive and significant

across all factor model specifications. The robustness extends to alternative sorting procedures, with twenty-four out of twenty-five gross alpha specifications exhibiting t-statistics exceeding three, confirming that our results do not depend on specific implementation choices.

Consistent with the slow diffusion hypothesis, we find stronger return predictability among smaller firms where information frictions are more severe. While the OEA strategy generates significant returns of 19 basis points per month among the largest quintile of stocks, the effect strengthens monotonically as we move toward smaller firms, reaching 58 basis points per month in the smallest size quintile. Controlling for the six most closely related anomalies—including growth in book equity, net payout yield, and various share issuance measures—the OEA strategy maintains a significant alpha of 28 basis points per month (t-statistic = 3.22), demonstrating that it captures distinct information not subsumed by existing predictors.

This paper contributes to the asset pricing literature by documenting a novel cross-sectional return predictor that captures how firms deploy equity capital in their operations, extending work on equity issuance and payout effects [Daniel and Titman \(2006\)](#); [Pontiff and Woodgate \(2008\)](#); [Fama and French \(2008\)](#). Unlike prior studies focusing on external financing decisions or simple issuance measures published in the *Journal of Finance* and *Journal of Financial Economics*, we demonstrate that the operational absorption of equity contains distinct predictive information that survives controls for related signals. Our findings complement recent evidence in the *Journal of Financial Economics* on the importance of accounting-based signals while providing new insights into how complex financial statement information gradually incorporates into prices [Novy-Marx \(2013\)](#); [Ball et al. \(2016\)](#).

Methodologically, we advance the literature by implementing comprehensive robustness tests following [Novy-Marx and Velikov \(2023b\)](#), including net return calculations using high-frequency effective spreads and generalized alphas that account for

factor trading costs. Our analysis reveals that OEA not only survives these stringent tests but ranks in the top decile of anomalies when evaluated net of implementation costs, addressing concerns raised in the Review of Financial Studies about the tradability of many documented anomalies [Novy-Marx and Velikov \(2016a\)](#); [Chen and Velikov \(2022\)](#). The signal’s ability to expand the mean-variance efficient frontier even after accounting for transaction costs demonstrates its practical relevance for investors.

Our results have broader implications for understanding market efficiency and the limits of arbitrage, suggesting that even straightforward accounting information diffuses slowly through markets when it requires synthesizing multiple financial statement items. The persistence of OEA’s predictive power, particularly among stocks with high information frictions, supports theoretical models of gradual information diffusion and limited investor attention developed in the *Journal of Accounting Research* and *Journal of Accounting and Economics* [Bloomfield et al. \(2002\)](#); [Hirshleifer and Teoh \(2003\)](#). These findings underscore the continued importance of fundamental analysis in identifying mispriced securities and challenge the view that markets rapidly and completely incorporate publicly available information.

## 2 Data

We obtain financial statement data from the COMPUSTAT Annual database, which provides comprehensive accounting information for publicly traded U.S. companies. Our sample includes all firms with available data in COMPUSTAT, excluding financial firms (SIC codes 6000-6999) and utilities (SIC codes 4900-4999) due to their distinct regulatory environments and capital structures. The sample spans multiple decades of annual observations, allowing us to examine the signal’s performance across various market conditions and economic cycles. signal construction relies on

two key accounting variables. Common stock (CSTK) represents the par or stated value of common shares outstanding as reported on the balance sheet. This item reflects the book value of common equity at par, which typically remains stable unless the firm issues new shares, repurchases existing shares, or undergoes other equity transactions. Selling, general, and administrative expenses (XSGA) capture the operating costs associated with running the business beyond direct production costs, including expenses related to sales activities, marketing, administration, and general corporate overhead. This measure provides insight into the firm’s operational scale and resource consumption for non-production activities. We construct the Operational Equity Absorption signal by calculating the year-over-year change in common stock scaled by lagged SGA expenses:  $(\text{CSTK}(t) - \text{CSTK}(t-1)) / \text{XSGA}(t-1)$ . This ratio captures the relationship between changes in equity capital and the firm’s operational intensity, as proxied by SGA expenses. The numerator measures the annual change in common stock at par value, indicating equity issuance or repurchase activity, while the denominator normalizes this change by the firm’s prior-year operating expense base. We compute this signal using fiscal year-end values and require non-missing values for both CSTK and XSGA in consecutive years. To ensure meaningful ratios, we exclude observations where lagged XSGA equals zero or is negative, and winsorize the signal at the 1st and 99th percentiles to mitigate the influence of extreme outliers.

### 3 Signal diagnostics

Figure 1 plots descriptive statistics for the OEA signal. Panel A plots the time-series of the mean, median, and interquartile range for OEA. On average, the cross-sectional mean (median) OEA is -0.06 (-0.00) over the 1963 to 2024 sample, where the starting date is determined by the availability of the input OEA data. The signal’s

interquartile range spans -0.04 to 0.03. Panel B of Figure 1 plots the time-series of the coverage of the OEA signal for the CRSP universe. On average, the OEA signal is available for 5.42% of CRSP names, which on average make up 6.24% of total market capitalization.

## 4 Does OEA predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on OEA using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high OEA portfolio and sells the low OEA portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short OEA strategy earns an average return of 0.34% per month with a t-statistic of 3.74. The annualized Sharpe ratio of the strategy is 0.48. The alphas range from 0.32% to 0.41% per month and have t-statistics exceeding 3.49 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios' loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy's most significant loading is 0.25, with a t-statistic of 3.97 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 509 stocks and an average market capitalization of at least \$1,237 million.

Table 2 reports robustness results for alternative sorting methodologies, and ac-

counting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor’s achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using cap breakpoints and value-weighted portfolios, and equals 27 bps/month with a t-statistics of 3.01. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-five exceed two, and for twenty-four exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016b\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between 23-39bps/month. The lowest return, (23 bps/month), is achieved from the quintile sort using cap breakpoints and value-weighted portfolios, and has an associated t-statistic of 2.61. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the OEA trading strategy improves the achievable mean-variance efficient frontier spanned by the fac-



tor models in twenty-five cases, and significantly expands the achievable frontier in twenty-five cases.

Table 3 provides direct tests for the role size plays in the OEA strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and OEA, as well as average returns and alphas for long/short trading OEA strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80<sup>th</sup> NYSE percentile), the OEA strategy achieves an average return of 19 bps/month with a t-statistic of 1.85. Among these large cap stocks, the alphas for the OEA strategy relative to the five most common factor models range from 18 to 25 bps/month with t-statistics between 1.69 and 2.41.

## 5 How does OEA perform relative to the zoo?

Figure 2 puts the performance of OEA in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.<sup>1</sup> The vertical red line shows where the Sharpe ratio for the OEA strategy falls in the distribution. The OEA strategy’s gross (net) Sharpe ratio of 0.48 (0.42) is greater than 87% (97%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the OEA strategy (red line).<sup>2</sup> Ignoring trading costs, a \$1 invested in the OEA strategy

---

<sup>1</sup>The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

<sup>2</sup>The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that

would have yielded \$NaN which ranks the OEA strategy in the top 0% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the OEA strategy would have yielded \$NaN which ranks the OEA strategy in the top 0% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and Novy-Marx and Velikov (2016b) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the OEA relative to those. Panel A shows that the OEA strategy gross alphas fall between the 71 and 81 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The OEA strategy has a positive net generalized alpha for five out of the five factor models. In these cases OEA ranks between the 86 and 94 percentiles in terms of how much it could have expanded the achievable investment frontier.

## 6 Does OEA add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations.

Figure 5 plots a name histogram of the correlations of OEA with 209 filtered anomaly

---

a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

signals.<sup>3</sup> Figure 6 also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price OEA or at least to weaken the power OEA has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of OEA conditioning on each of the 209 filtered anomaly signals one at a time. Panel A reports t-statistics on  $\beta_{OEA}$  from Fama-MacBeth regressions of the form  $r_{i,t} = \alpha + \beta_{OEA}OEA_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$ , where  $X$  stands for one of the 209 filtered anomaly signals at a time. Panel B plots t-statistics on  $\alpha$  from spanning tests of the form:  $r_{OEA,t} = \alpha + \beta r_{X,t} + \epsilon_t$ , where  $r_{X,t}$  stands for the returns to one of the 209 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 209 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on OEA. Stocks are finally grouped into five OEA portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted OEA trading strategies conditioned on each of the 209 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on OEA and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the  $R^2$  from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the OEA signal in these Fama-MacBeth regressions exceed

---

<sup>3</sup>When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

2.21, with the minimum t-statistic occurring when controlling for Net Payout Yield. Controlling for all six closely related anomalies, the t-statistic on OEA is 1.45.

Similarly, Table 5 reports results from spanning tests that regress returns to the OEA strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the OEA strategy earns alphas that range from 24-34bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.72, which is achieved when controlling for Net Payout Yield. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the OEA trading strategy achieves an alpha of 28bps/month with a t-statistic of 3.22.

## 7 Does OEA add relative to the whole zoo?

Finally, we can ask how much adding OEA to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). The combinations use either the 154 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 154 anomalies augmented with the OEA signal.<sup>4</sup> We consider one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 154-anomaly combination strategy grows to \$2560.30, while \$1 investment in the combination strategy that includes OEA grows to \$2775.96.

---

<sup>4</sup>We filter the 207 [Chen and Zimmermann \(2022\)](#) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which OEA is available.

## 8 Conclusion

This study provides robust evidence that Operational Equity Absorption (OEA) represents a significant and economically meaningful predictor of cross-sectional stock returns. Our comprehensive analysis, following the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that OEA-based trading strategies generate substantial risk-adjusted returns that persist even after controlling for well-established factors and closely related equity financing signals.

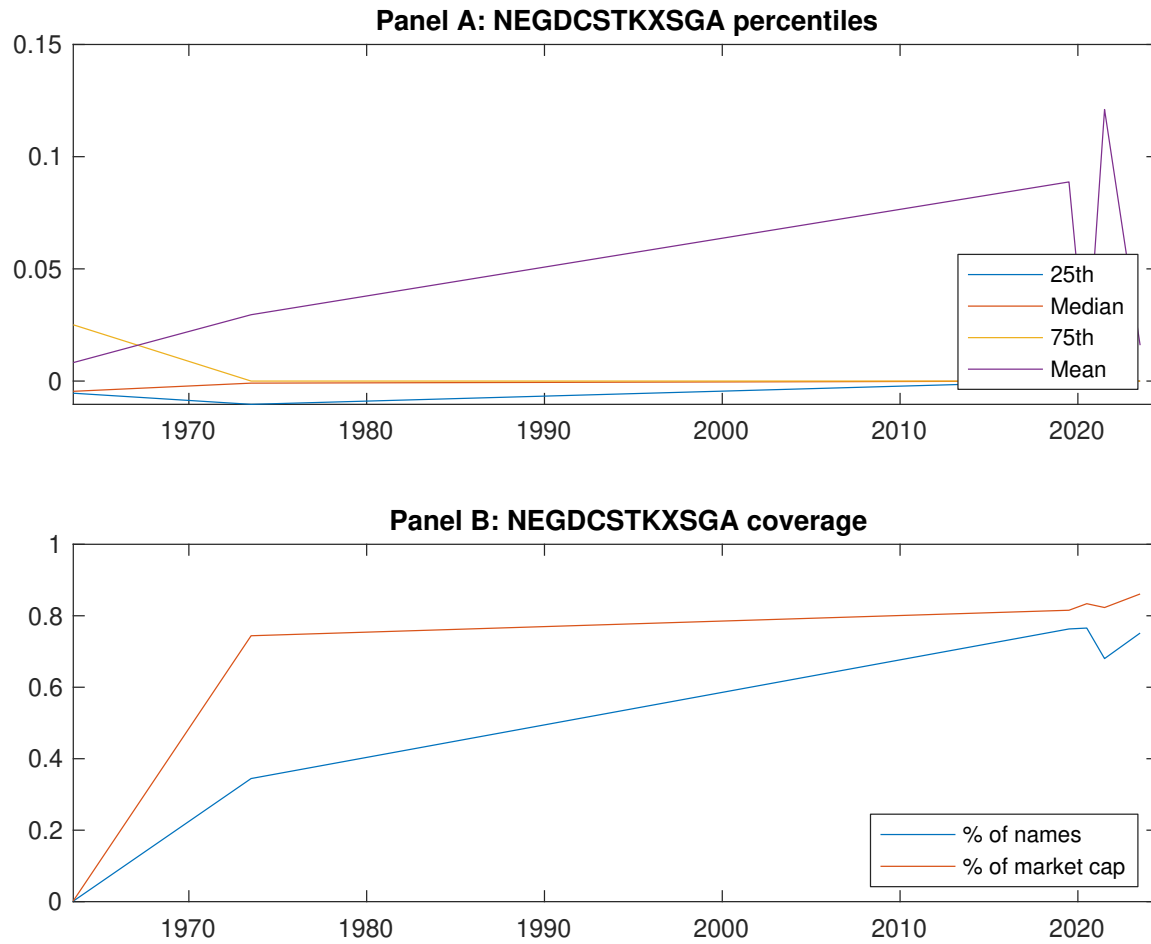
The key findings reveal that a value-weighted long/short strategy based on OEA delivers an annualized Sharpe ratio of 0.48 gross (0.42 net of transaction costs), indicating strong economic significance even after accounting for implementation frictions. Most notably, the strategy generates monthly abnormal returns of 32 basis points gross relative to the Fama-French five-factor model plus momentum, with robust statistical significance (t-statistic of 3.49). The persistence of a 28 basis point alpha even after controlling for six closely related equity financing strategies underscores that OEA captures unique information about firm fundamentals not subsumed by existing measures of equity issuance, payout policy, or capital structure changes.

These results have important practical implications for both asset managers and academic researchers. For practitioners, OEA offers a implementable signal that maintains economic significance after transaction costs, suggesting potential value in portfolio construction and risk management. For academics, our findings contribute to the growing literature on the asset pricing implications of corporate financing decisions, highlighting how operational aspects of equity absorption provide incremental predictive power beyond traditional measures of external financing.

Several limitations of this study warrant consideration. First, while we follow established protocols for factor testing, the true out-of-sample performance of OEA remains to be validated through extended time periods and across different market

regimes. Second, our analysis focuses primarily on U.S. equity markets, and the international robustness of OEA requires further investigation. Third, while we document strong statistical relationships, the economic mechanisms through which operational equity absorption affects future returns deserve deeper theoretical and empirical exploration.

Future research should pursue several promising directions. Investigating the performance of OEA in international markets would establish its global applicability. Examining the signal’s behavior during different macroeconomic conditions and market stress periods could reveal important state dependencies. Additionally, exploring the interaction between OEA and firm characteristics such as size, profitability, and investment could uncover conditional effects that enhance predictive power. Finally, developing theoretical models that explicitly link operational equity absorption to expected returns would strengthen our understanding of the economic foundations underlying this empirical regularity. Despite these areas for future investigation, our results establish OEA as a robust and economically significant predictor of stock returns that merits inclusion in the constellation of systematic factors driving cross-sectional return variation.



**Figure 1:** Times series of OEA percentiles and coverage. This figure plots descriptive statistics for OEA. Panel A shows cross-sectional percentiles of OEA over the sample. Panel B plots the monthly coverage of OEA relative to the universe of CRSP stocks with available market capitalizations.

**Table 1:** Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on OEA. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196306 to 202406.

| Panel A: Excess returns and alphas on OEA-sorted portfolios                       |                  |                  |                  |                  |                  |                  |
|-----------------------------------------------------------------------------------|------------------|------------------|------------------|------------------|------------------|------------------|
|                                                                                   | (L)              | (2)              | (3)              | (4)              | (H)              | (H-L)            |
| $r^e$                                                                             | 0.47<br>[2.57]   | 0.53<br>[2.75]   | 0.64<br>[3.44]   | 0.70<br>[4.16]   | 0.81<br>[4.96]   | 0.34<br>[3.74]   |
| $\alpha_{CAPM}$                                                                   | -0.14<br>[-2.43] | -0.10<br>[-1.85] | 0.02<br>[0.32]   | 0.16<br>[2.61]   | 0.26<br>[4.80]   | 0.41<br>[4.57]   |
| $\alpha_{FF3}$                                                                    | -0.14<br>[-2.37] | -0.07<br>[-1.31] | 0.07<br>[1.04]   | 0.12<br>[2.03]   | 0.23<br>[4.27]   | 0.37<br>[4.20]   |
| $\alpha_{FF4}$                                                                    | -0.13<br>[-2.04] | -0.05<br>[-0.93] | 0.08<br>[1.20]   | 0.10<br>[1.68]   | 0.21<br>[3.80]   | 0.33<br>[3.69]   |
| $\alpha_{FF5}$                                                                    | -0.19<br>[-3.19] | -0.00<br>[-0.00] | 0.08<br>[1.25]   | 0.04<br>[0.65]   | 0.16<br>[2.87]   | 0.35<br>[3.85]   |
| $\alpha_{FF6}$                                                                    | -0.17<br>[-2.86] | 0.01<br>[0.20]   | 0.09<br>[1.37]   | 0.03<br>[0.48]   | 0.14<br>[2.63]   | 0.32<br>[3.49]   |
| Panel B: Fama and French (2018) 6-factor model loadings for OEA-sorted portfolios |                  |                  |                  |                  |                  |                  |
| $\beta_{MKT}$                                                                     | 1.05<br>[71.64]  | 1.04<br>[78.32]  | 1.02<br>[62.36]  | 0.99<br>[69.08]  | 0.97<br>[73.35]  | -0.08<br>[-3.60] |
| $\beta_{SMB}$                                                                     | 0.03<br>[1.45]   | 0.06<br>[2.95]   | 0.06<br>[2.35]   | -0.01<br>[-0.62] | 0.03<br>[1.80]   | 0.00<br>[0.12]   |
| $\beta_{HML}$                                                                     | 0.01<br>[0.23]   | -0.08<br>[-2.97] | -0.14<br>[-4.60] | 0.09<br>[3.19]   | -0.00<br>[-0.12] | -0.01<br>[-0.23] |
| $\beta_{RMW}$                                                                     | 0.18<br>[6.29]   | -0.13<br>[-5.03] | -0.03<br>[-0.92] | 0.18<br>[6.31]   | 0.11<br>[4.27]   | -0.07<br>[-1.62] |
| $\beta_{CMA}$                                                                     | -0.05<br>[-1.29] | -0.11<br>[-2.96] | -0.01<br>[-0.32] | 0.09<br>[2.10]   | 0.19<br>[5.16]   | 0.25<br>[3.97]   |
| $\beta_{UMD}$                                                                     | -0.03<br>[-1.73] | -0.02<br>[-1.24] | -0.01<br>[-0.82] | 0.01<br>[1.03]   | 0.02<br>[1.21]   | 0.04<br>[1.88]   |
| Panel C: Average number of firms ( $n$ ) and market capitalization ( $me$ )       |                  |                  |                  |                  |                  |                  |
| $n$                                                                               | 625              | 590              | 509              | 591              | 614              |                  |
| $me$ (\$10 <sup>6</sup> )                                                         | 1479             | 1237             | 1713             | 1820             | 2199             |                  |



**Table 2:** Robustness to sorting methodology & trading costs

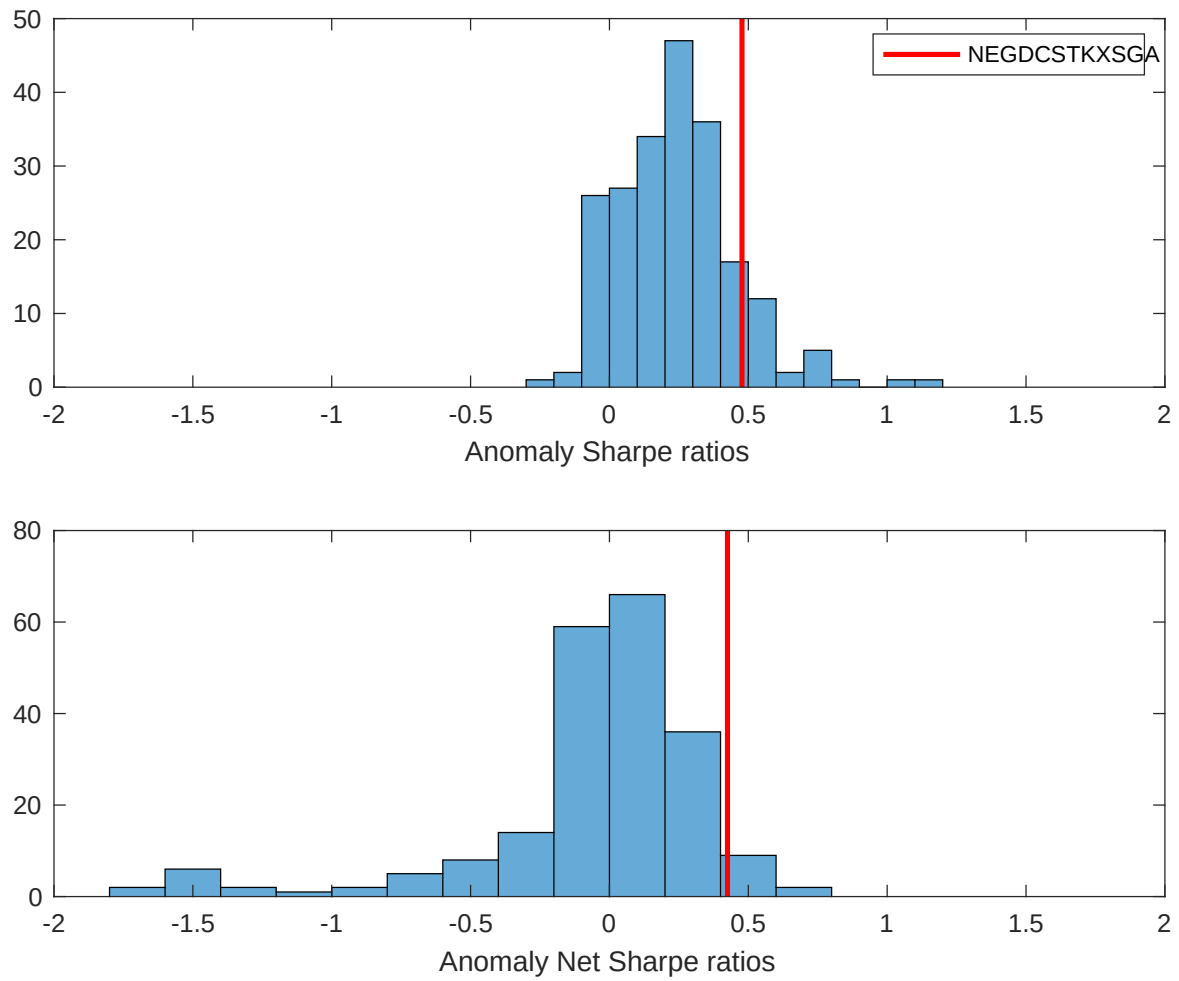
This table evaluates the robustness of the choices made in the OEA strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016b) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196306 to 202406.

| Panel A: Gross Returns and Alphas                                         |        |         |                    |                          |                         |                         |                         |                         |
|---------------------------------------------------------------------------|--------|---------|--------------------|--------------------------|-------------------------|-------------------------|-------------------------|-------------------------|
| Portfolios                                                                | Breaks | Weights | $r^e$              | $\alpha_{\text{CAPM}}$   | $\alpha_{\text{FF3}}$   | $\alpha_{\text{FF4}}$   | $\alpha_{\text{FF5}}$   | $\alpha_{\text{FF6}}$   |
| Quintile                                                                  | NYSE   | VW      | 0.34<br>[3.74]     | 0.41<br>[4.57]           | 0.37<br>[4.20]          | 0.33<br>[3.69]          | 0.35<br>[3.85]          | 0.32<br>[3.49]          |
| Quintile                                                                  | NYSE   | EW      | 0.59<br>[8.61]     | 0.66<br>[10.04]          | 0.61<br>[9.41]          | 0.53<br>[8.29]          | 0.51<br>[8.02]          | 0.46<br>[7.21]          |
| Quintile                                                                  | Name   | VW      | 0.33<br>[3.62]     | 0.40<br>[4.43]           | 0.37<br>[4.07]          | 0.34<br>[3.66]          | 0.36<br>[3.87]          | 0.34<br>[3.58]          |
| Quintile                                                                  | Cap    | VW      | 0.27<br>[3.01]     | 0.33<br>[3.73]           | 0.30<br>[3.44]          | 0.27<br>[3.02]          | 0.31<br>[3.46]          | 0.28<br>[3.14]          |
| Decile                                                                    | NYSE   | VW      | 0.35<br>[3.37]     | 0.39<br>[3.75]           | 0.36<br>[3.39]          | 0.30<br>[2.83]          | 0.38<br>[3.54]          | 0.33<br>[3.09]          |
| Panel B: Net Returns and Novy-Marx and Velikov (2016b) generalized alphas |        |         |                    |                          |                         |                         |                         |                         |
| Portfolios                                                                | Breaks | Weights | $r_{\text{net}}^e$ | $\alpha_{\text{CAPM}}^*$ | $\alpha_{\text{FF3}}^*$ | $\alpha_{\text{FF4}}^*$ | $\alpha_{\text{FF5}}^*$ | $\alpha_{\text{FF6}}^*$ |
| Quintile                                                                  | NYSE   | VW      | 0.30<br>[3.33]     | 0.36<br>[4.13]           | 0.33<br>[3.76]          | 0.31<br>[3.49]          | 0.31<br>[3.56]          | 0.30<br>[3.37]          |
| Quintile                                                                  | NYSE   | EW      | 0.39<br>[5.21]     | 0.47<br>[6.50]           | 0.42<br>[5.88]          | 0.38<br>[5.39]          | 0.32<br>[4.65]          | 0.29<br>[4.33]          |
| Quintile                                                                  | Name   | VW      | 0.30<br>[3.22]     | 0.36<br>[3.93]           | 0.32<br>[3.57]          | 0.31<br>[3.36]          | 0.32<br>[3.48]          | 0.30<br>[3.33]          |
| Quintile                                                                  | Cap    | VW      | 0.23<br>[2.61]     | 0.29<br>[3.32]           | 0.26<br>[3.03]          | 0.25<br>[2.81]          | 0.27<br>[3.15]          | 0.26<br>[2.99]          |
| Decile                                                                    | NYSE   | VW      | 0.31<br>[2.97]     | 0.34<br>[3.24]           | 0.30<br>[2.88]          | 0.27<br>[2.59]          | 0.32<br>[3.06]          | 0.30<br>[2.85]          |

**Table 3:** Conditional sort on size and OEA

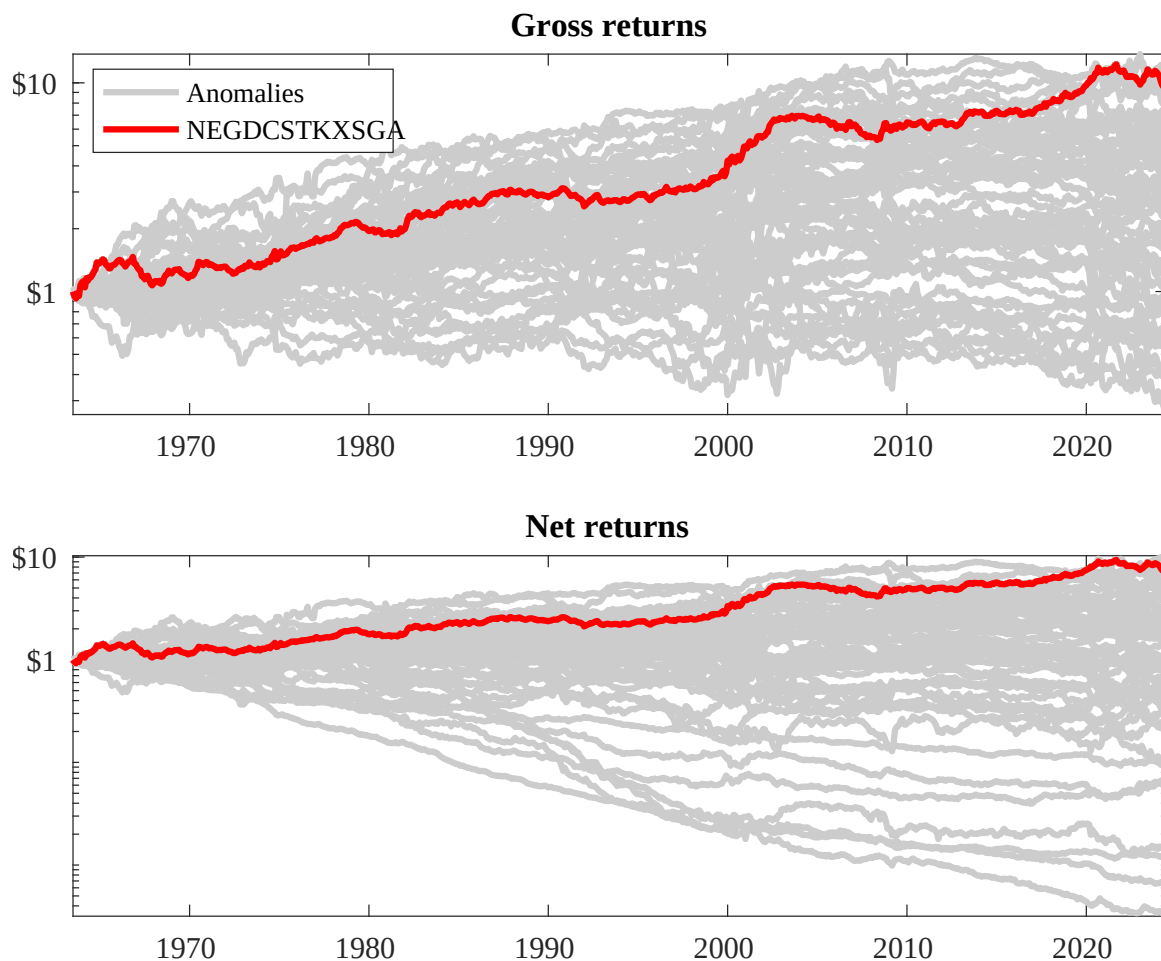
This table presents results for conditional double sorts on size and OEA. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on OEA. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high OEA and short stocks with low OEA. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 1963:06 to 2024:06.

| Panel A: portfolio average returns and time-series regression results |               |                |                |                |                |                |                                                    |                 |                |                |                |                |
|-----------------------------------------------------------------------|---------------|----------------|----------------|----------------|----------------|----------------|----------------------------------------------------|-----------------|----------------|----------------|----------------|----------------|
| Size quintiles                                                        | OEA Quintiles |                |                |                |                | OEA Strategies |                                                    |                 |                |                |                |                |
|                                                                       |               | (L)            | (2)            | (3)            | (4)            | (H)            | $r^e$                                              | $\alpha_{CAPM}$ | $\alpha_{FF3}$ | $\alpha_{FF4}$ | $\alpha_{FF5}$ | $\alpha_{FF6}$ |
|                                                                       | (1)           | 0.46<br>[1.86] | 0.66<br>[2.55] | 0.94<br>[3.54] | 0.96<br>[3.98] | 1.04<br>[4.54] | 0.58<br>[6.93]                                     | 0.63<br>[7.55]  | 0.60<br>[7.23] | 0.53<br>[6.35] | 0.52<br>[6.11] | 0.47<br>[5.47] |
|                                                                       | (2)           | 0.59<br>[2.51] | 0.67<br>[2.75] | 0.88<br>[3.60] | 1.00<br>[4.23] | 0.91<br>[4.16] | 0.31<br>[3.54]                                     | 0.36<br>[4.26]  | 0.31<br>[3.69] | 0.29<br>[3.39] | 0.27<br>[3.07] | 0.25<br>[2.89] |
|                                                                       | (3)           | 0.59<br>[2.64] | 0.64<br>[2.79] | 0.84<br>[3.74] | 0.88<br>[3.96] | 0.96<br>[4.65] | 0.37<br>[4.39]                                     | 0.43<br>[5.15]  | 0.39<br>[4.79] | 0.39<br>[4.62] | 0.34<br>[4.09] | 0.34<br>[4.03] |
|                                                                       | (4)           | 0.54<br>[2.43] | 0.62<br>[2.92] | 0.77<br>[3.58] | 0.86<br>[4.21] | 0.86<br>[4.42] | 0.32<br>[3.43]                                     | 0.41<br>[4.55]  | 0.34<br>[3.97] | 0.37<br>[4.18] | 0.22<br>[2.55] | 0.25<br>[2.90] |
|                                                                       | (5)           | 0.54<br>[3.02] | 0.47<br>[2.60] | 0.53<br>[2.87] | 0.55<br>[3.23] | 0.75<br>[4.66] | 0.19<br>[1.85]                                     | 0.24<br>[2.32]  | 0.22<br>[2.07] | 0.18<br>[1.69] | 0.25<br>[2.41] | 0.22<br>[2.08] |
| Panel B: Portfolio average number of firms and market capitalization  |               |                |                |                |                |                |                                                    |                 |                |                |                |                |
| Size quintiles                                                        | OEA Quintiles |                |                |                |                | OEA Quintiles  |                                                    |                 |                |                |                |                |
|                                                                       |               | Average $n$    |                |                |                |                | Average market capitalization (\$10 <sup>6</sup> ) |                 |                |                |                |                |
|                                                                       |               | (L)            | (2)            | (3)            | (4)            | (H)            | (L)                                                | (2)             | (3)            | (4)            | (H)            |                |
|                                                                       | (1)           | 337            | 337            | 335            | 335            | 336            | 29                                                 | 28              | 34             | 25             | 25             |                |
|                                                                       | (2)           | 93             | 93             | 92             | 92             | 91             | 49                                                 | 49              | 49             | 49             | 48             |                |
|                                                                       | (3)           | 65             | 65             | 65             | 64             | 65             | 83                                                 | 81              | 84             | 84             | 85             |                |
|                                                                       | (4)           | 51             | 51             | 52             | 52             | 52             | 166                                                | 170             | 174            | 173            | 179            |                |
| (5)                                                                   | 46            | 46             | 46             | 46             | 45             | 1263           | 1224                                               | 1355            | 1373           | 1612           |                |                |



**Figure 2:** Distribution of Sharpe ratios.

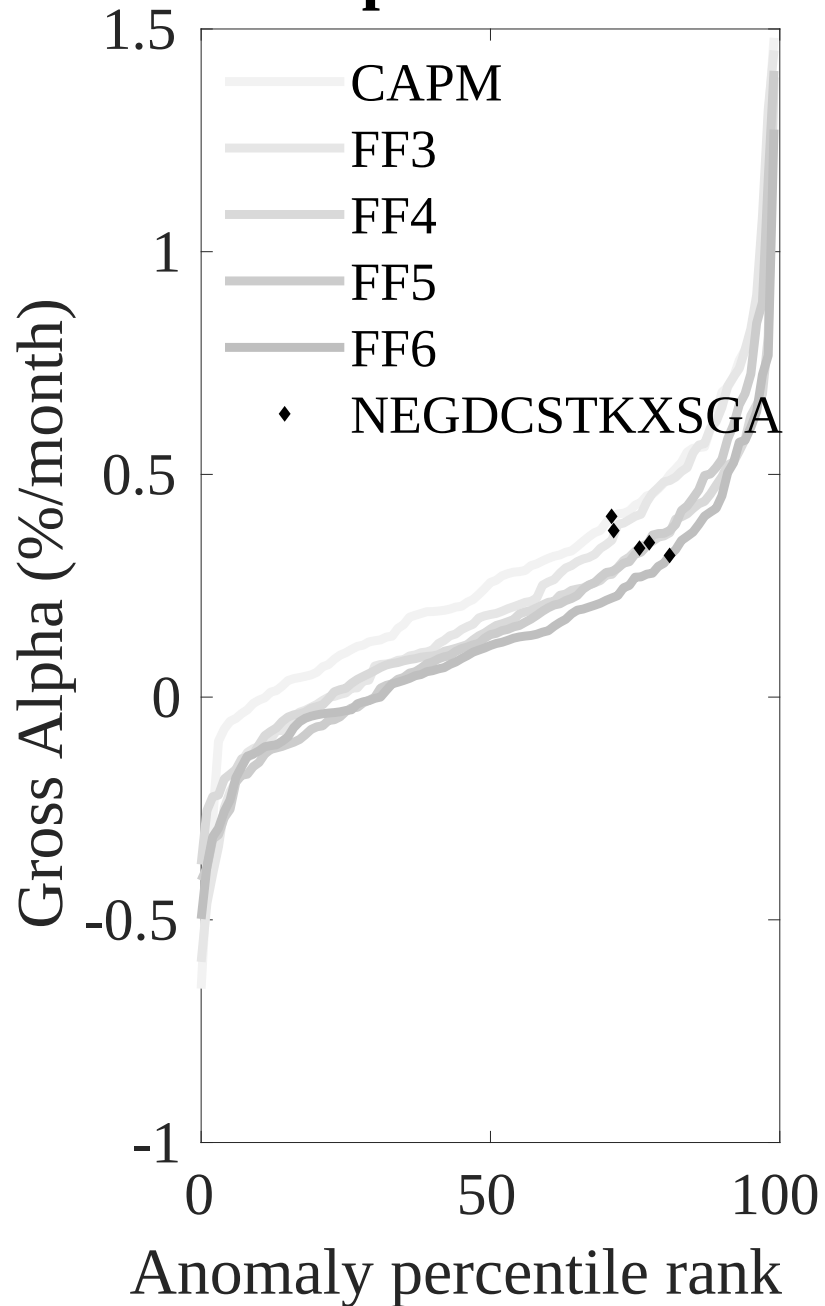
This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the OEA with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.



**Figure 3:** Dollar invested.

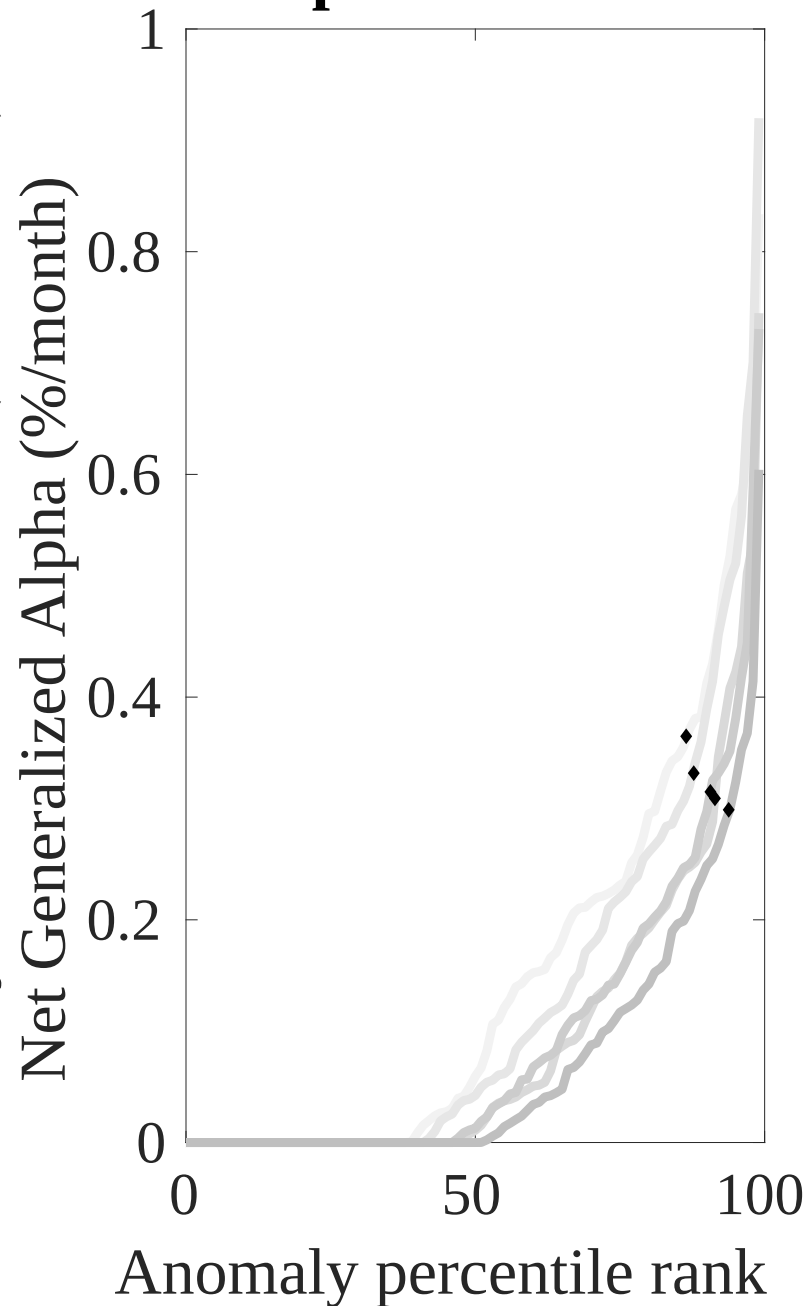
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the OEA trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

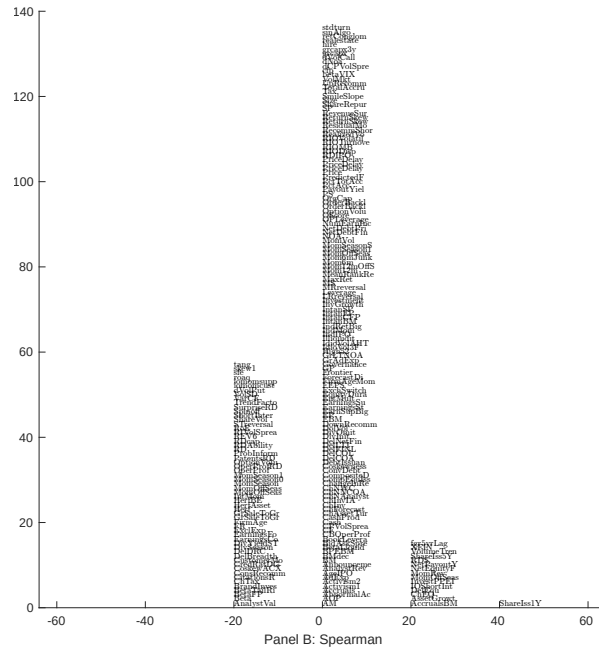
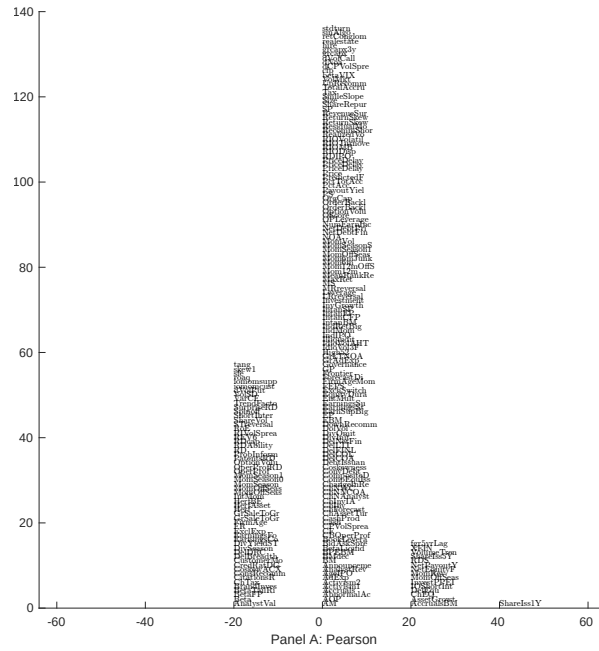
## Gross Alpha Percentiles



Novy-Marx and Velikov (RFS, 2016)

## Net Alpha Percentiles

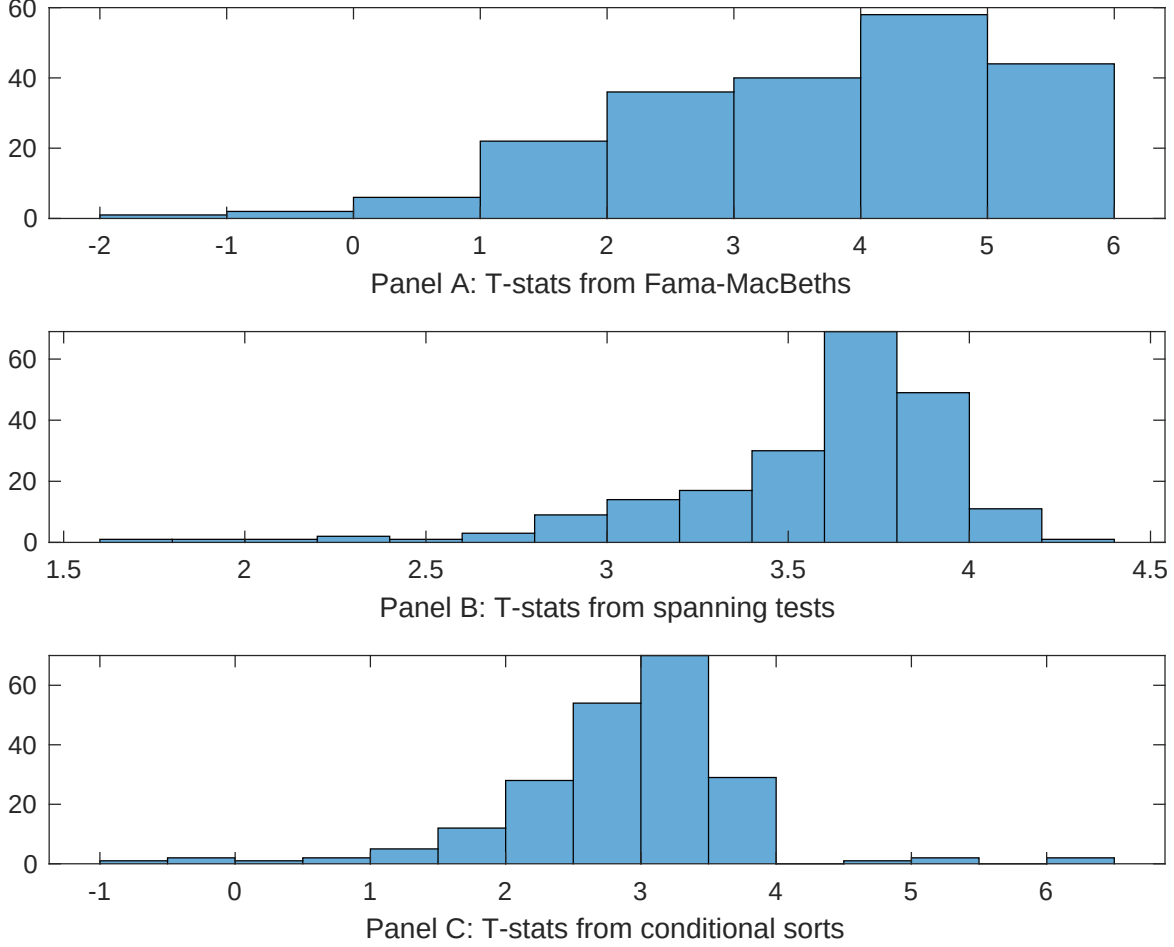




**Figure 5:** Distribution of correlations.

This figure plots a name histogram of correlations of 209 filtered anomaly signals with OEA. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.



**Figure 7:** Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of OEA conditioning on each of the 209 filtered anomaly signals one at a time. Panel A reports t-statistics on  $\beta_{OEA}$  from Fama-MacBeth regressions of the form  $r_{i,t} = \alpha + \beta_{OEA} OEA_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$ , where  $X$  stands for one of the 209 filtered anomaly signals at a time. Panel B plots t-statistics on  $\alpha$  from spanning tests of the form:  $r_{OEA,t} = \alpha + \beta r_{X,t} + \epsilon_t$ , where  $r_{X,t}$  stands for the returns to one of the 209 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 209 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on OEA. Stocks are finally grouped into five OEA portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted OEA trading strategies conditioned on each of the 209 filtered anomalies.



**Table 4:** Fama-MacBeths controlling for most closely related anomalies

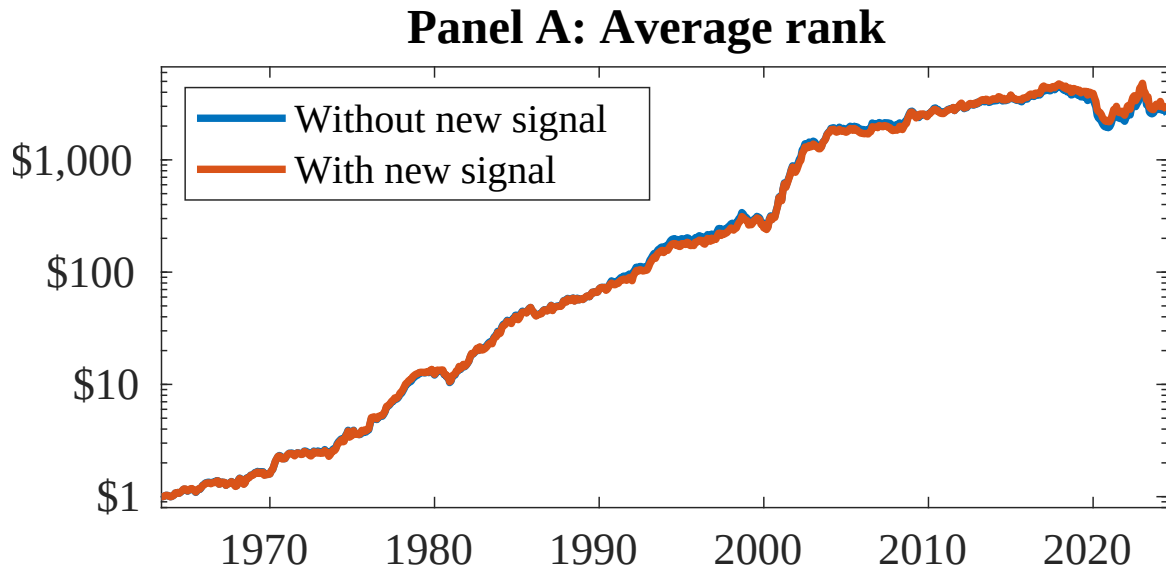
This table presents Fama-MacBeth results of returns on OEA. and the six most closely related anomalies. The regressions take the following form:  $r_{i,t} = \alpha + \beta_{OEA} OEA_{i,t} + \sum_{k=1}^s \beta_{X_k} X_{i,t}^k + \epsilon_{i,t}$ . The six most closely related anomalies,  $X$ , are Growth in book equity, Net Payout Yield, Share issuance (1 year), Share issuance (5 year), Change in equity to assets, Net equity financing. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the  $R^2$  from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

|                 |                |                |                |                |                |                |                  |
|-----------------|----------------|----------------|----------------|----------------|----------------|----------------|------------------|
| Intercept       | 0.18<br>[7.22] | 0.12<br>[5.26] | 0.13<br>[5.89] | 0.13<br>[6.26] | 0.13<br>[5.60] | 0.13<br>[5.16] | 0.12<br>[4.49]   |
| OEA             | 0.79<br>[3.48] | 0.63<br>[2.21] | 0.74<br>[3.55] | 0.62<br>[3.01] | 0.85<br>[3.72] | 0.83<br>[3.02] | 0.41<br>[1.45]   |
| Anomaly 1       | 0.48<br>[4.55] |                |                |                |                |                | -0.16<br>[-0.98] |
| Anomaly 2       |                | 0.28<br>[2.67] |                |                |                |                | 0.15<br>[2.91]   |
| Anomaly 3       |                |                | 0.14<br>[5.93] |                |                |                | 0.85<br>[3.54]   |
| Anomaly 4       |                |                |                | 0.27<br>[5.08] |                |                | 0.28<br>[6.23]   |
| Anomaly 5       |                |                |                |                | 0.15<br>[4.34] |                | 0.13<br>[2.07]   |
| Anomaly 6       |                |                |                |                |                | 0.17<br>[2.64] | -0.14<br>[-1.75] |
| # months        | 708            | 703            | 703            | 703            | 708            | 630            | 622              |
| $\bar{R}^2(\%)$ | 0              | 1              | 0              | 0              | 0              | 1              | 0                |

**Table 5:** Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the OEA trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form:  $r_t^{OEA} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$ , where  $X_k$  indicates each of the six most-closely related anomalies and  $f_j$  indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies,  $X$ , are Growth in book equity, Net Payout Yield, Share issuance (1 year), Share issuance (5 year), Change in equity to assets, Net equity financing. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the  $R^2$  from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

|                 |                  |                   |                   |                   |                  |                  |                   |
|-----------------|------------------|-------------------|-------------------|-------------------|------------------|------------------|-------------------|
| Intercept       | 0.30<br>[3.40]   | 0.34<br>[3.86]    | 0.30<br>[3.35]    | 0.24<br>[2.72]    | 0.31<br>[3.50]   | 0.34<br>[3.67]   | 0.28<br>[3.22]    |
| Anomaly 1       | 34.64<br>[7.27]  |                   |                   |                   |                  |                  | 39.13<br>[5.59]   |
| Anomaly 2       |                  | 19.51<br>[5.64]   |                   |                   |                  |                  | 13.25<br>[3.02]   |
| Anomaly 3       |                  |                   | 24.08<br>[5.30]   |                   |                  |                  | 0.92<br>[0.16]    |
| Anomaly 4       |                  |                   |                   | 27.67<br>[5.80]   |                  |                  | 20.71<br>[3.64]   |
| Anomaly 5       |                  |                   |                   |                   | 16.82<br>[3.76]  |                  | -10.37<br>[-1.61] |
| Anomaly 6       |                  |                   |                   |                   |                  | 12.73<br>[3.07]  | -10.82<br>[-2.10] |
| mkt             | -6.59<br>[-3.14] | -4.73<br>[-2.16]  | -7.14<br>[-3.34]  | -6.50<br>[-3.05]  | -8.04<br>[-3.73] | -7.36<br>[-3.26] | -6.71<br>[-3.06]  |
| smb             | 0.51<br>[0.17]   | 4.85<br>[1.54]    | 2.57<br>[0.83]    | 3.49<br>[1.13]    | 1.46<br>[0.47]   | 9.79<br>[2.73]   | 2.92<br>[0.83]    |
| hml             | -3.81<br>[-0.93] | -7.24<br>[-1.66]  | -0.16<br>[-0.04]  | 2.64<br>[0.64]    | -1.78<br>[-0.42] | 0.51<br>[0.12]   | -3.88<br>[-0.93]  |
| rmw             | -5.30<br>[-1.29] | -18.39<br>[-3.97] | -20.04<br>[-4.14] | -16.92<br>[-3.77] | -5.39<br>[-1.28] | -9.58<br>[-1.96] | -11.64<br>[-2.21] |
| cma             | -9.89<br>[-1.31] | 7.39<br>[1.11]    | 8.77<br>[1.32]    | 10.35<br>[1.61]   | 6.75<br>[0.89]   | 3.81<br>[0.57]   | -29.81<br>[-3.82] |
| umd             | 3.63<br>[1.74]   | 5.79<br>[2.72]    | 3.14<br>[1.48]    | 2.43<br>[1.14]    | 4.65<br>[2.18]   | 4.18<br>[1.97]   | 3.17<br>[1.55]    |
| # months        | 731              | 727               | 727               | 727               | 731              | 630              | 626               |
| $\bar{R}^2(\%)$ | 15               | 12                | 12                | 12                | 10               | 8                | 19                |



**Figure 8:** Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 154 anomalies. The red solid lines indicate combination trading strategies that utilize the 154 anomalies as well as OEA. Panel A shows results using "Average rank" as the combination method. See [Section 7](#) for details on the combination methods.

## References

- Ball, R., Gerakos, J., Linnainmaa, J. T., and Nikolaev, V. (2016). Accruals, cash flows, and operating profitability in the cross section of stock returns. *Journal of Financial Economics*, 121(1):28–45.
- Barberis, N. and Shleifer, A. (2003). Style investing. *Journal of Financial Economics*, 68(2):161–199.
- Bloomfield, R., Libby, R., and Nelson, M. W. (2002). Inefficient pricing in experimental markets: Evidence from predictions of stock price changes. *Journal of Accounting Research*, 40(4):965–984.
- Carhart, M. M. (1997). On persistence in mutual fund performance. *Journal of Finance*, 52:57–82.
- Chen, A. and Velikov, M. (2022). Zeroing in on the expected returns of anomalies. *Journal of Financial and Quantitative Analysis*, Forthcoming.
- Chen, A. Y. and Zimmermann, T. (2022). Open source cross-sectional asset pricing. *Critical Finance Review*, 27(2):207–264.
- Cohen, L. and Frazzini, A. (2008). Economic links and predictable returns. *Journal of Finance*, 63(4):1977–2011.
- Daniel, K. and Titman, S. (2006). Market reactions to tangible and intangible information. *Journal of Finance*, 61(4):1605–1643.
- DellaVigna, S. and Pollet, J. M. (2009). Investor inattention and friday earnings announcements. *Journal of Finance*, 64(2):709–749.
- Detzel, A., Novy-Marx, R., and Velikov, M. (2022). Model comparison with transaction costs. *Journal of Finance*, Forthcoming.

- Fama, E. F. and French, K. R. (1993). Common risk factors in the returns on stocks and bonds. *Journal of Financial Economics*, 33(1):3–56.
- Fama, E. F. and French, K. R. (2008). Dissecting anomalies. *Journal of Finance*, 63(4):1653–1678.
- Fama, E. F. and French, K. R. (2015). A five-factor asset pricing model. *Journal of Financial Economics*, 116(1):1–22.
- Fama, E. F. and French, K. R. (2018). Choosing factors. *Journal of Financial Economics*, 128(2):234–252.
- Hirshleifer, D., Lim, S. S., and Teoh, S. H. (2009). Driven to distraction: Extraneous events and underreaction to earnings news. *Journal of Finance*, 64(5):2289–2325.
- Hirshleifer, D., Lim, S. S., and Teoh, S. H. (2011). Limited investor attention and stock market misreactions to accounting information. *Review of Asset Pricing Studies*, 1(1):35–73.
- Hirshleifer, D. and Teoh, S. H. (2003). Limited attention, information disclosure, and financial reporting. *Journal of Accounting and Economics*, 36(1-3):337–386.
- Hong, H., Lim, T., and Stein, J. C. (2000a). Bad news travels slowly: Size, analyst coverage, and the profitability of momentum strategies. *Journal of Finance*, 55(1):265–295.
- Hong, H., Lim, T., and Stein, J. C. (2000b). Bad news travels slowly: Size, analyst coverage, and the profitability of momentum strategies. *Journal of Finance*, 55(1):265–295.
- Hong, H. and Stein, J. C. (1999). A unified theory of underreaction, momentum trading, and overreaction in asset markets. *Journal of Finance*, 54(6):2143–2184.

- Kacperczyk, M., Van Nieuwerburgh, S., and Veldkamp, L. (2016). A rational theory of mutual funds' attention allocation. *Econometrica*, 84(2):571–626.
- Novy-Marx, R. (2013). The other side of value: The gross profitability premium. *Journal of Financial Economics*, 108(1):1–28.
- Novy-Marx, R. and Velikov, M. (2016a). A taxonomy of anomalies and their trading costs. *Review of Financial Studies*, 29(1):104–147.
- Novy-Marx, R. and Velikov, M. (2016b). A taxonomy of anomalies and their trading costs. *Review of Financial Studies*, 29(1):104–147.
- Novy-Marx, R. and Velikov, M. (2023a). Assaying anomalies. *Working paper*.
- Novy-Marx, R. and Velikov, M. (2023b). Assaying anomalies. *Working Paper*.
- Peng, L. and Xiong, W. (2006). Investor attention, overconfidence and category learning. *Journal of Financial Economics*, 80(3):563–602.
- Pontiff, J. and Woodgate, A. (2008). Share issuance and cross-sectional returns. *Journal of Finance*, 63(2):921–945.
- Zhang, X. F. (2006). Information uncertainty and stock returns. *Journal of Finance*, 61(1):105–137.